

VIETNAM GENERAL CONFEDERATION OF LABOUR

TON DUC THANG UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY



GROUP 11

**ENDTERM REPORT ON
DATA MINING**

**FIRE AND SMOKE DETECTION
USING COMPUTER VISION**

HO CHI MINH CITY, 2026

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Advised by

Dr. Hoang Anh

FIRE AND SMOKE DETECTION

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Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
AP	Average Precision
COCO	Common Objects in Context annotation format
CV	Computer Vision
dHash	Difference perceptual hash
EDA	Exploratory Data Analysis
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
IoU	Intersection over Union
mAP	mean Average Precision
NMS	Non-Maximum Suppression
QA	Quality Assurance
SAHI	Slicing Aided Hyper Inference
TP	True Positive
YOLO	You Only Look Once object detector family

FIRE AND SMOKE DETECTION USING COMPUTER VISION

Executive Summary

This report presents a complete Data Mining and Computer Vision pipeline for detecting fire and smoke in diverse Vietnamese visual contexts. The project does not simply fine-tune an existing detector on a clean benchmark. Instead, the main contribution is the construction of a traceable end-to-end workflow: noisy multi-source data collection, custom segmentation labeling, label quality control, duplicate and near-duplicate removal, group-aware dataset design, train-only slicing for rare small-object cases, YOLO11x detection training, and systematic error analysis.

The final model direction is a single-class YOLO11x detector named `fire_smoke`. Although the original labels are built as segmentation masks, the deployment-oriented model is formulated as detection because an alerting system needs fast and stable bounding-box outputs more than pixel-perfect masks. The training was conducted on rented NVIDIA L4 GPU infrastructure, with an experimental cost of approximately 1,000,000 VND. This cost produced real checkpoints, training curves, validation metrics, full-image test metrics, sliced inference ablations, qualitative prediction samples, and a local Streamlit interface for testing inference with the packaged Stage 1/2 model artifact.

The best Stage 2 validation result reached **mAP@50 = 0.454** and **mAP@50:95 = 0.262**. On the original full-image test split, the same checkpoint achieved **Precision = 0.552**, **Recall = 0.430**, **mAP@50 = 0.421**, and **mAP@50:95 = 0.240**. These metrics are below the original target of $\text{mAP@50} \geq 0.70$, but they are substantially more informative than the earlier sliced-only result. The sliced inference ablation reached only $\text{mAP@50} = 0.152$, showing that the current slicing and merging protocol is not yet tuned. The report therefore emphasizes an academically honest conclusion: the current model is not deployment-ready, but the dataset, pipeline, and error analysis establish a strong foundation for active learning, relabeling, hard-negative expansion, and threshold/NMS optimization.

Chapter 1

Introduction

1.1 Problem Context

Early fire and smoke detection is a safety-critical Computer Vision problem. The objective is to detect visual indicators of fire or smoke from still images or camera streams before an incident escalates. In contrast to many conventional object detection tasks, fire and smoke do not have stable shapes, rigid contours, or consistent texture. Fire changes rapidly over time, smoke is semi-transparent, and early-stage ignition can occupy only a tiny area of the frame.

This project focuses on contexts that are visually relevant to Vietnam: residential alleyways, dense urban backgrounds, low-light footage, small flame spots, smoke mixed with haze, and hard negative scenes such as lamps, candles, reflections, and warm-colored lighting. These scenarios are deliberately harder than clean forest-fire or large-flame examples.

1.2 Why This is a Data Mining Project

The central problem is not only model selection. A high-capacity detector cannot perform well if the data are noisy, duplicated, mislabeled, or evaluation splits leak near-identical frames. The project therefore follows the Data Mining lifecycle:

1. Collect raw visual data from heterogeneous sources.
2. Organize data into meaningful semantic groups.
3. Build segmentation annotations where no original masks existed.
4. Audit labels and remove corrupted or irrelevant samples.
5. Detect exact and perceptual duplicates.
6. Build deterministic train/validation/test splits.
7. Train YOLO11x using a two-stage strategy.

8. Analyze metrics by dataset group and failure mode.

1.3 Key Contributions

The report should be evaluated not only by the final mAP value, but also by the engineering work required to produce a valid experimental setting. The main contributions are:

- **Multi-source data collection:** Images were collected and selected from Kaggle, Google Images, self-curated Vietnamese contexts, hard negatives, and ambient null scenes.
- **Custom segmentation labeling:** The original public data did not provide segmentation masks. Fire and smoke regions were manually annotated and quality checked.
- **Data quality control:** The pipeline detects corrupted files, empty labels, invalid coordinates, exact duplicates, and near-duplicate frames.
- **Leakage-aware splitting:** Splits are performed at source-image level before slicing to avoid inflated validation scores.
- **Compute-backed training:** The YOLO11x detector was trained and fine-tuned on rented NVIDIA L4 GPU resources, costing approximately 1,000,000 VND.
- **Honest error analysis:** The report explains why the metrics are below the original target and identifies concrete improvement directions.

1.4 System Objective

The original technical target was to approach $\text{mAP}@50 \geq 0.70$ for a one-class fire/smoke detector. The achieved result is lower, but the system still satisfies several important research and engineering objectives:

- a reproducible dataset preparation pipeline;
- a curated dataset with 13,896 images and 28,279 annotated objects after deduplication;
- a two-stage YOLO11x training workflow;
- validation, full-image test, per-group, and sliced inference evaluation artifacts;
- a published Hugging Face model artifact and a local Streamlit inference UI for checkpoint testing;
- a clear diagnosis of false negatives, hard-negative false positives, and sliced inference weaknesses.

Chapter 2

Data Collection and Annotation

2.1 Data Sources

The dataset was designed to avoid a common failure in fire detection research: training on obvious large fires and then failing on early-stage or ambiguous camera scenes. Table 2.1 summarizes the data sources.

Table 2.1: Data sources and their roles

Source	Contribution	Main Issues
Kaggle images	Diverse public fire/smoke examples.	Images only; no segmentation masks.
Google Images	Civilian scenes, small fire, faint smoke, local contexts.	Watermarks, duplicates, inconsistent quality.
Self-curated data	Vietnamese alleyways and residential contexts.	Requires normalization and manual labeling.
Hard negatives	Lamps, candles, reflections, warm lighting, non-fire smoke-like patterns.	Must remain unlabeled to teach rejection.
Ambient null images	Normal background frames without incidents.	Important for reducing false alarms.

2.2 Dataset Availability

The prepared dataset is published on Hugging Face at:

<https://huggingface.co/datasets/thanhhoangnvbg/fire-vn-yolo11seg-v1>

The submission package does not include the local dataset files because they are large and can be regenerated from the public dataset repository using the provided preparation scripts.

This keeps the submitted archive focused on the report, code, and reproducible processing pipeline.

2.3 Model Artifact and Inference Interface

The Stage 1 and Stage 2 training outputs are packaged separately from the source code because checkpoint files are large. The model artifact is published on Hugging Face at:

<https://huggingface.co/thanhhoangnvbg/fire-detection-yolo11-stage12>

The downloadable artifact is `model_artifacts_stage12_complete.tar.gz`. After extraction, the default inference checkpoint is the Stage 2 fine-tuned model:

```
weights/fire-detection-yolo11-stage12/runs/final/yolo11x_detect_fire_smoke_14_
finetune/weights/best.pt
```

For practical testing, the project includes a Streamlit interface in `app.py`. The interface loads the checkpoint, accepts uploaded images or bundled sample images, allows confidence, IoU, image-size, and device settings to be adjusted, and displays both the original image and the annotated prediction output. This UI is intended as a reproducible inference test tool, not as a production monitoring system.

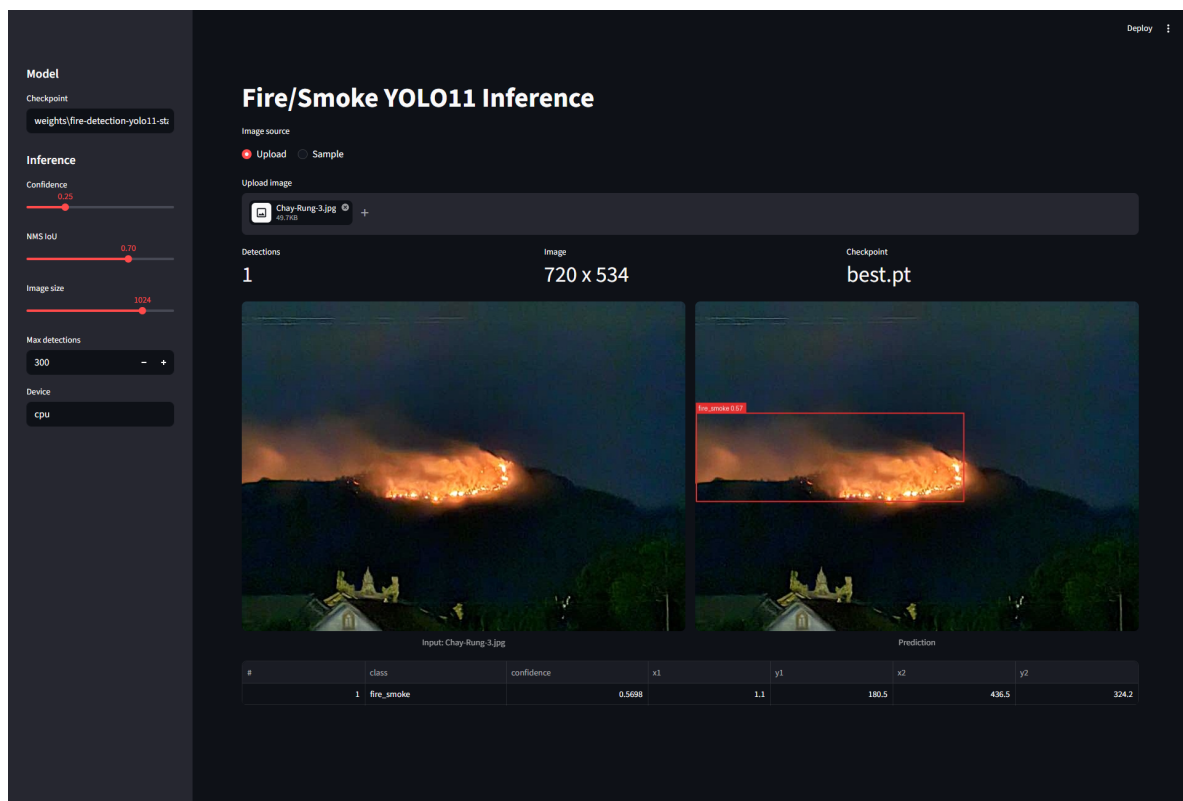


Figure 2.1: Streamlit inference interface for testing the packaged Stage 2 YOLO11 checkpoint on Chay-Rung-3 . jpg.

2.4 Dataset Group Design

Instead of treating all images as a flat pool, the dataset is organized into five groups. This design makes it possible to diagnose exactly where the model succeeds or fails.

Table 2.2: Dataset groups

Group	Name	Role
g01	positive_standard	Main positive data with visible fire/smoke instances.
g02	alley_context	Difficult Vietnamese alley/residential scenes.
g03	hard_negative	Non-fire confusing samples such as lamps and reflections.
g04	sahi_small_objects	Small fire/smoke objects that require special slicing.
g05	ambient_null	Normal background images used to reduce false alarms.

2.5 Manual Segmentation Annotation

A major effort of the project is the creation of segmentation annotations. Many source images had no pixel-level labels. The team therefore annotated visible fire and smoke regions with polygon masks. This is much more expensive than image-level classification because annotators must decide where the flame or smoke boundary begins and ends.

The annotation protocol follows these principles:

- label only visible fire/smoke regions, not the entire bright area;
- do not label watermarks, timestamps, camera overlays, or unrelated text;
- label partially occluded fire only where it is visible;
- label smoke only when it has enough visual structure to distinguish it from background haze;
- keep hard negatives unlabeled even if they contain fire-like colors;
- export verified data to COCO segmentation and then convert it to YOLO formats.

2.6 Why the Labeling Work Matters

The final metric is not high, but the labeling work remains valuable. Without segmentation labels, the model would either learn only image-level correlations or rely on noisy bounding

boxes. Segmentation masks preserve shape information and allow a clean conversion into detection boxes. This makes the final one-class detector more grounded in actual fire/smoke regions.

The practical challenge is that fire and smoke are not rigid objects. Flame boundaries are fuzzy and smoke is semi-transparent. Too-wide masks make the model learn background; too-tight masks reduce recall on diffuse smoke and spreading fire. The labeling process therefore represents a trade-off between geometric precision and practical detection stability.

2.7 Annotation Quality Limitation

Although the report documents the annotation protocol, the current artifact set does not contain a quantitative annotation-quality study. This is a real limitation. With six members participating in data collection and labeling, label inconsistency is a plausible source of noise. Examples include different interpretations of faint smoke, inconsistent treatment of reflections, and different polygon tightness around flame boundaries.

The missing measurements are:

- **Inter-annotator agreement:** A shared subset should be labeled by at least two annotators and compared using mask IoU, box IoU, or object-level F1.
- **Label error rate:** A random audit sample should be reviewed after export to estimate missing labels, wrong labels, overly wide masks, and false positive labels.
- **Boundary consistency:** Smoke and flame masks should be evaluated separately because smoke has more subjective boundaries than flame.

Because these measurements were not completed, the report should not overclaim that the labels are fully consistent. A more accurate conclusion is that the labels were created and QA-checked procedurally, but their statistical reliability remains an open experimental issue.

Chapter 3

Exploratory Data Analysis

3.1 Final Dataset Overview

After preprocessing and near-duplicate removal, the final dataset contains 13,896 images and 28,279 annotated objects. Table 3.1 summarizes the final state.

Table 3.1: Final dataset overview after deduplication

Statistic	Value
Total images	13,896
Total segmentation objects	28,279
Background images	4,099
Background ratio	29.50%
Smoke objects	12,356
Fire objects	15,923

3.2 Before and After Deduplication

Deduplication intentionally reduces the apparent amount of data. This is not a weakness: it prevents optimistic metrics caused by nearly identical frames appearing across splits.

Table 3.2: Dataset size before and after near-duplicate removal

Stage	Images	Objects	Interpretation
Before near-dedupe	15,615	31,134	Detection-level dataset before removing near-duplicate scenes.
After near-dedupe	13,896	28,279	Final dataset used for training/evaluation.
Change	-1,719	-2,855	Reduces repeated scenes and duplicate object evidence.

At source-image level, perceptual deduplication detected 1,095 near-duplicate clusters and removed 1,649 source images. Most removals came from g01 `positive_standard`, where repeated frames from the same event were common.

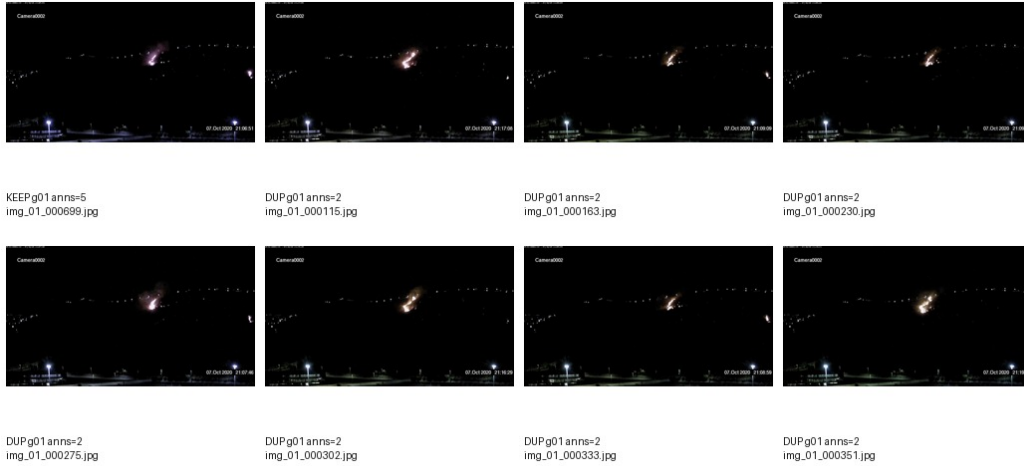


Figure 3.1: Example near-duplicate cluster found by perceptual hashing.

3.3 Split Distribution

The final train/validation/test split is shown in Table 3.3. Validation and test are kept as full images, while slicing is applied only to selected training groups.

Table 3.3: Distribution by split

Split	Images	Objects	Background Images
Train	12,322	25,086	3,628
Validation	841	1,724	252
Test	733	1,469	219

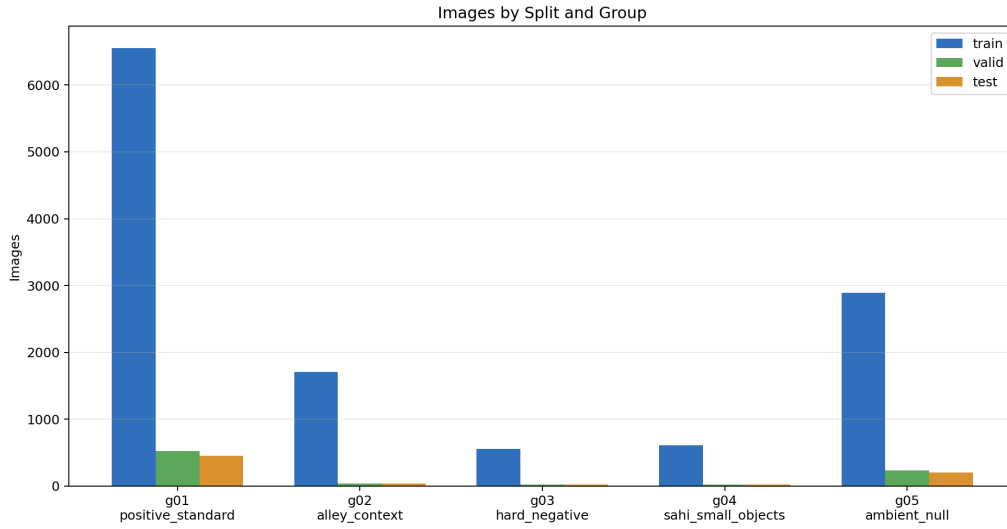


Figure 3.2: Images by split and group.

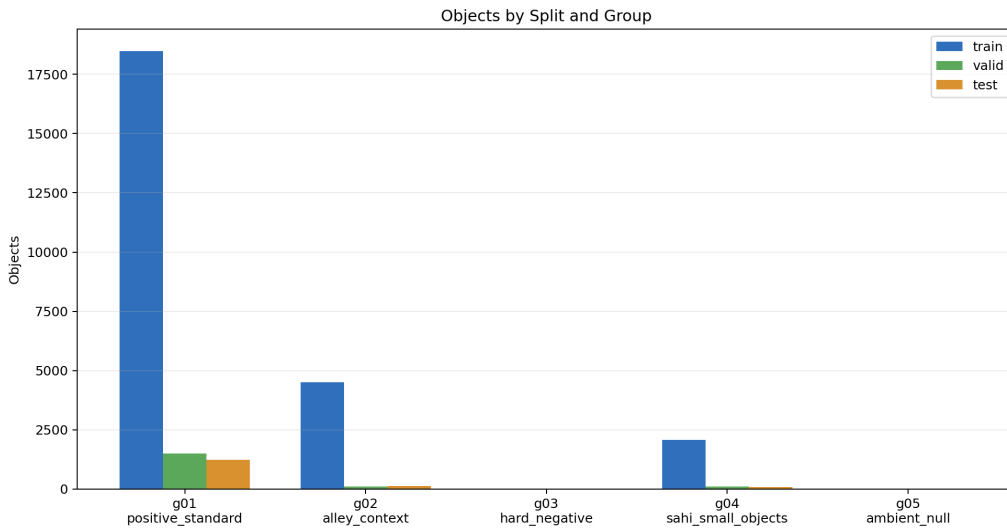


Figure 3.3: Objects by split and group.

3.4 Group Distribution

Table 3.4: Group-level split distribution

Split	Group	Images	Objects	Background Ratio
Train	g01 positive_standard	6,552	18,479	0.0000
Train	g02 alley_context	1,711	4,498	0.1099
Train	g03 hard_negative	555	0	1.0000
Train	g04 sahi_small_objects	612	2,086	0.0000

Split	Group	Images	Objects	Background Ratio
Train	g05 ambient_null	2,892	23	0.9976
Validation	g01 positive_standard	527	1,511	0.0000
Validation	g02 alley_context	42	105	0.0000
Validation	g03 hard_negative	19	0	1.0000
Validation	g04 sahi_small_objects	20	108	0.0000
Validation	g05 ambient_null	233	0	1.0000
Test	g01 positive_standard	452	1,245	0.0000
Test	g02 alley_context	42	132	0.0000
Test	g03 hard_negative	19	0	1.0000
Test	g04 sahi_small_objects	20	92	0.0000
Test	g05 ambient_null	200	0	1.0000

3.5 Visual and Statistical Patterns

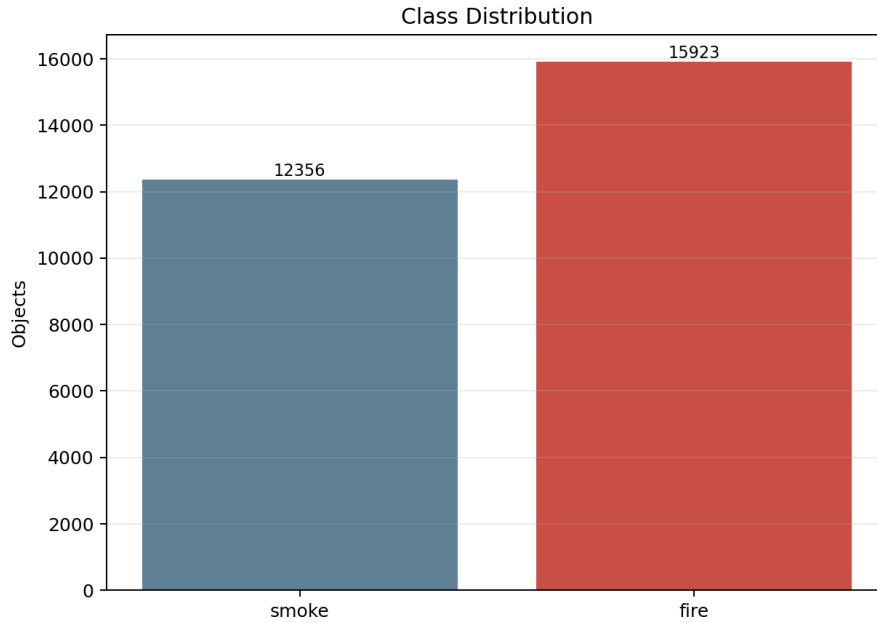


Figure 3.4: Class distribution of fire and smoke objects.

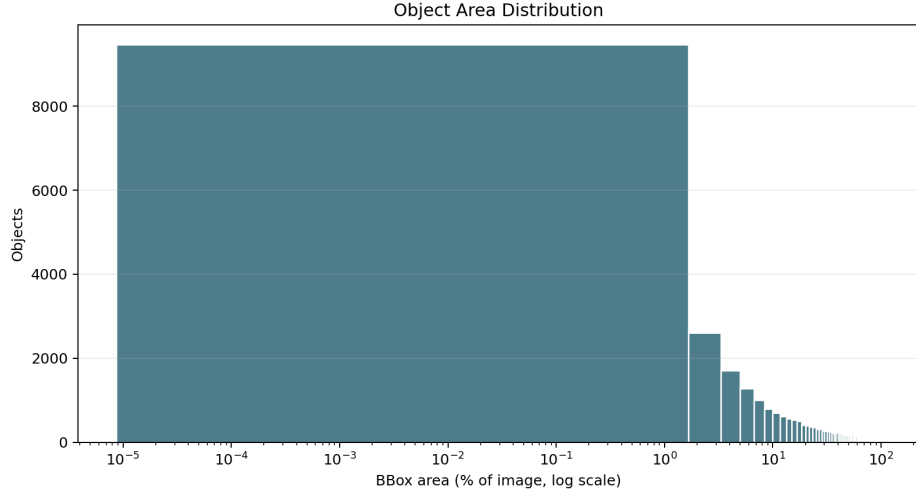


Figure 3.5: Object area distribution.

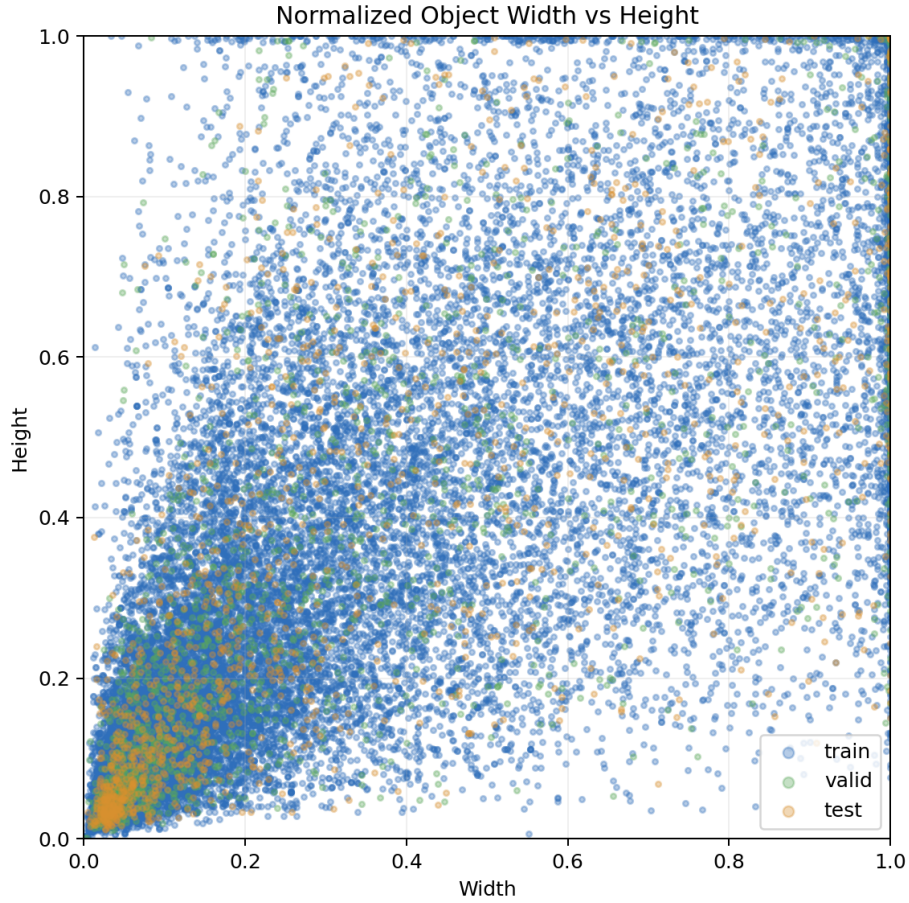


Figure 3.6: Object width-height scatter.

The EDA highlights three important points. First, g01 remains the dominant group, so its false negatives strongly influence the overall metric. Second, g03 and g05 contain no positive labels but are essential for reducing false alarms. Third, g04 has fewer images but a high object density,

making it important for early-stage small-fire detection.

Chapter 4

Data Preprocessing

4.1 Pipeline Overview

The preprocessing pipeline is implemented by project scripts such as `prepare_yolo11_seg_dataset.py`, `prepare_yolo11_detect_dataset.py`, and `generate_yolo11_data_report.py`. Figure 4.1 summarizes the workflow.

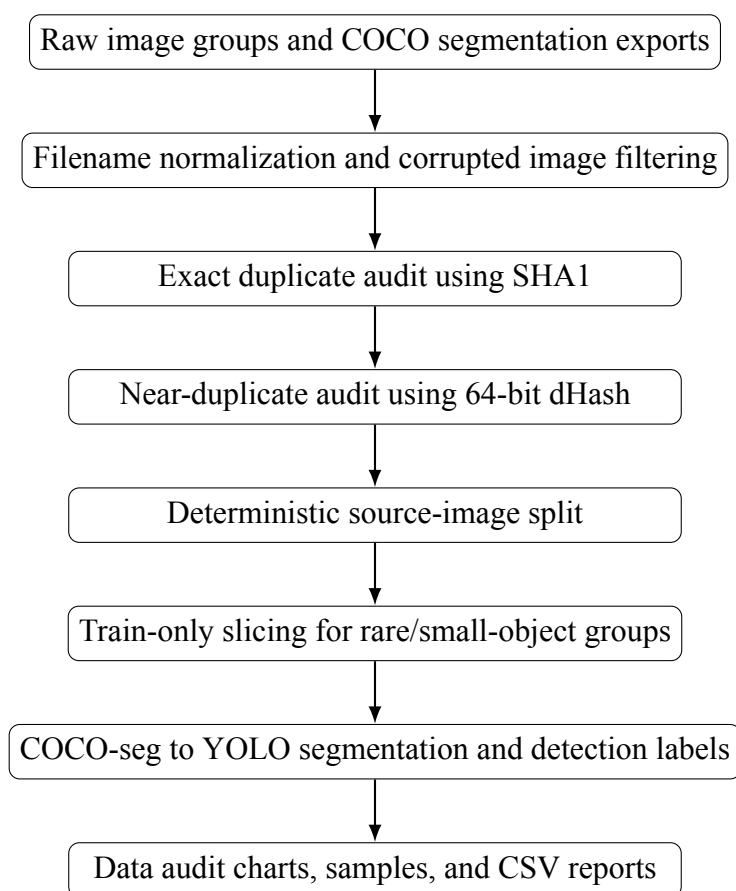


Figure 4.1: Data preprocessing pipeline.

4.2 Duplicate and Near-Duplicate Removal

Exact duplicates are detected by SHA1 file hashes. Near-duplicates are detected with a 64-bit dHash and Hamming distance threshold:

$$d_H(A, B) \leq 2.$$

This conservative threshold removes highly similar frames while limiting the risk of deleting useful visual variation.

Table 4.1: Near-duplicate removal by group

Group	Removed Source Images	Meaning
g01 positive_standard	1,518	Removes repeated fire scenes and video-frame sequences.
g02 alley_context	20	Preserves most rare alley-context images.
g03 hard_negative	9	Keeps hard negatives while removing obvious repetition.
g04 sahi_small_objects	0	Avoids losing rare small-object data.
g05 ambient_null	102	Reduces repeated normal background scenes.

4.3 Train-Only Slicing

Small fires can disappear after feature-map downsampling. Therefore, selected training groups are sliced into smaller tiles so that small objects occupy more pixels. Validation and test images are not sliced during dataset preparation, because evaluation should remain representative of real full-frame camera inputs.

Table 4.2: Training slicing configuration

Group	Slice Size	Overlap	Purpose
g02 alley_context	640	0.40	Increase rare alley-context samples.
g03 hard_negative	640	0.40	Keep hard background exposure with capped empty slices.
g04 sahi_small_objects	416	0.50	Enlarge small objects in training crops.

4.4 Why Preprocessing May Lower the Apparent Metric

Removing duplicates can reduce validation performance if the original split contained visually similar train/validation frames. This is desirable. A high score caused by data leakage is not useful for an alerting system. In safety-related applications, a lower but more honest score is better than an inflated metric.

Chapter 5

Model Design and Training Strategy

5.1 Why YOLO11x Detection

The project uses segmentation labels to build high-quality object boundaries, but the final model is detection-based. This decision is practical: fire/smoke alerting systems require fast inference, bounding boxes are sufficient for alarms, and detection metrics such as precision, recall, F1, and mAP@50 directly reflect the deployment objective.

Architecture of the Final YOLO11x Fire/Smoke Detection Pipeline

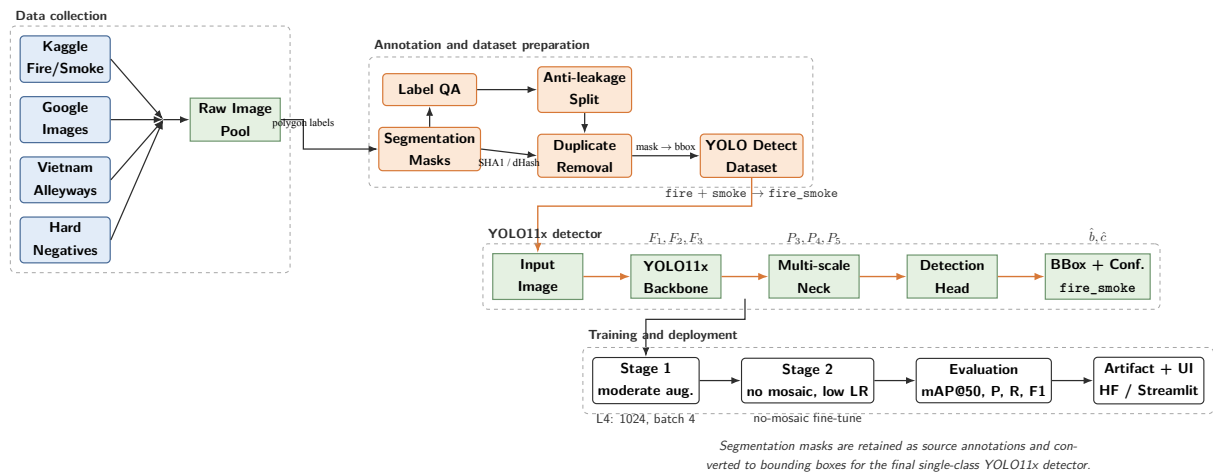


Figure 5.1: Final YOLO11x detection architecture and training pipeline.

5.2 Stage 1: Main Training

Stage 1 initializes from `yolo11x.pt`. It trains on the sliced training dataset and evaluates on full validation/test images. Its role is to produce a strong baseline checkpoint.

Table 5.1: Stage 1 training configuration

Parameter	Value
Model	YOLO11x detect
Image size	1024
Epochs	180
Batch size	4
Optimizer	AdamW
Initial learning rate	0.0005
Scheduler	Cosine LR
Warmup	4 epochs
Mosaic	0.40
MixUp	0.02
Scale	0.35

5.3 Stage 2: Fine-Tuning

Stage 2 starts from the Stage 1 best checkpoint and disables mosaic/mixup. This encourages the model to adapt to real full-image distributions rather than synthetic mosaic compositions.

The observed improvement is small. Stage 1 reaches $\text{mAP}@50 = 0.445$, while Stage 2 reaches $\text{mAP}@50 = 0.454$, an absolute gain of only 0.009. The best Stage 2 checkpoint appears at epoch 7 out of the planned 60 epochs, suggesting early plateau. Therefore, Stage 2 should be interpreted as a minor refinement rather than a major modeling breakthrough. It may still be useful because it slightly improves $\text{mAP}@50:95$ and stabilizes no-mosaic behavior, but the current evidence is not strong enough to claim that the fine-tuning design is optimal.

This also suggests two follow-up experiments:

1. compare Stage 2 against simply selecting the Stage 1 checkpoint with threshold tuning;
2. sweep fine-tuning learning rates and shorter epoch schedules, because the current setup may be spending unnecessary compute after early convergence.

Table 5.2: Stage 2 fine-tuning configuration

Parameter	Value
Initial weights	Stage 1 best checkpoint
Image size	1024
Epochs	60
Batch size	4
Learning rate	0.00008
Warmup	1 epoch
Mosaic	0.0
MixUp	0.0
Scale	0.20

5.4 Training Results by Stage

The Stage 2 checkpoint improves validation mAP slightly, but the model plateaus around $\text{mAP@50} = 0.45$. Table 5.3 summarizes the best validation metrics.

Table 5.3: Best validation metrics of Stage 1 and Stage 2

Stage	Best Epoch	Precision	Recall	mAP@50	mAP@50:95
Stage 1	97	0.578	0.432	0.445	0.253
Stage 2	7	0.582	0.430	0.454	0.262

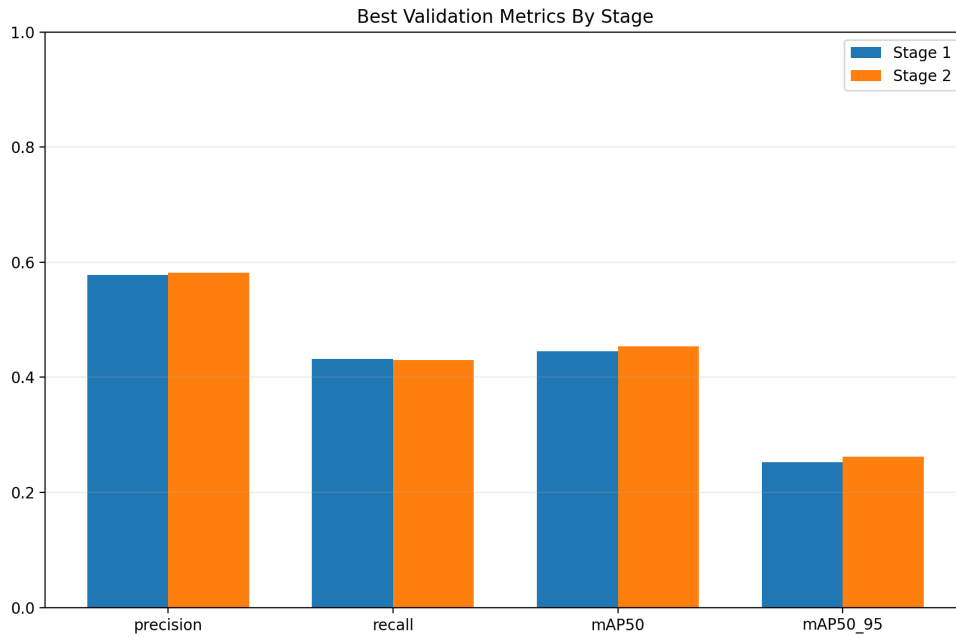


Figure 5.2: Comparison of best Stage 1 and Stage 2 metrics.

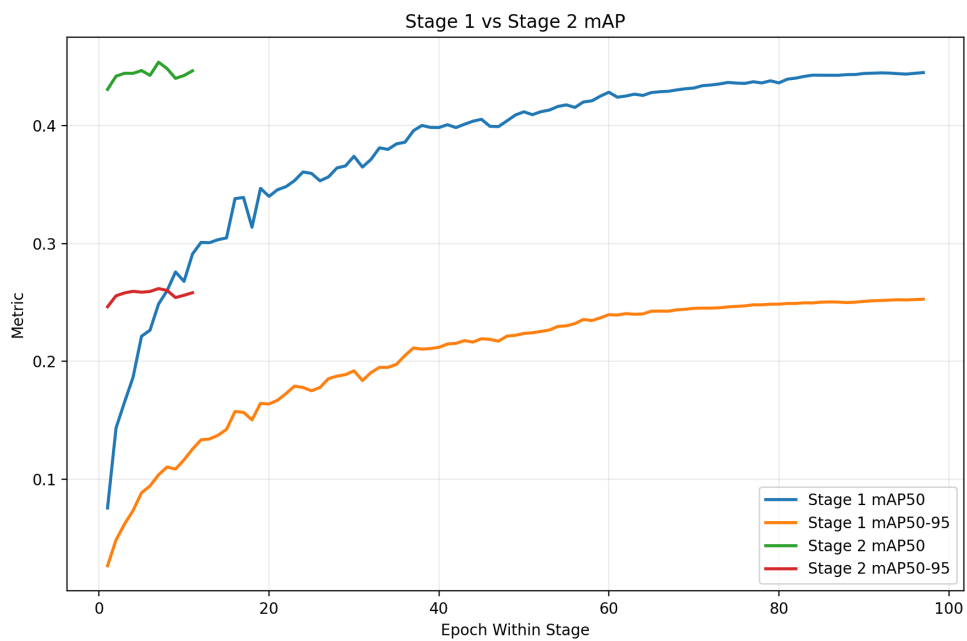


Figure 5.3: Stage 1 and Stage 2 mAP comparison.

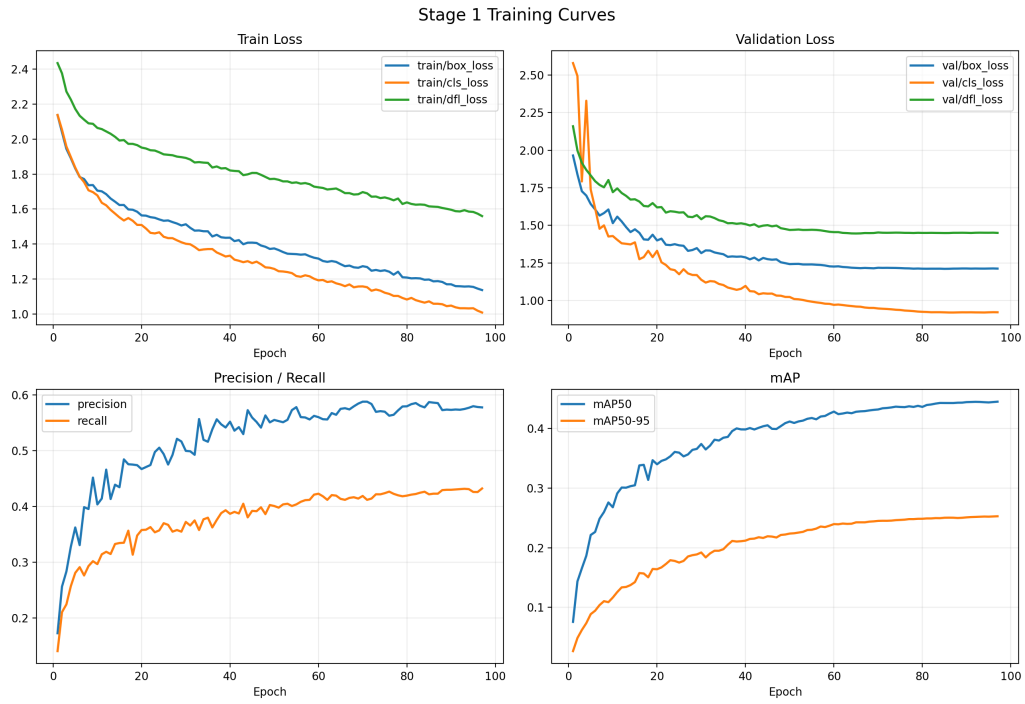


Figure 5.4: Stage 1 training curves.

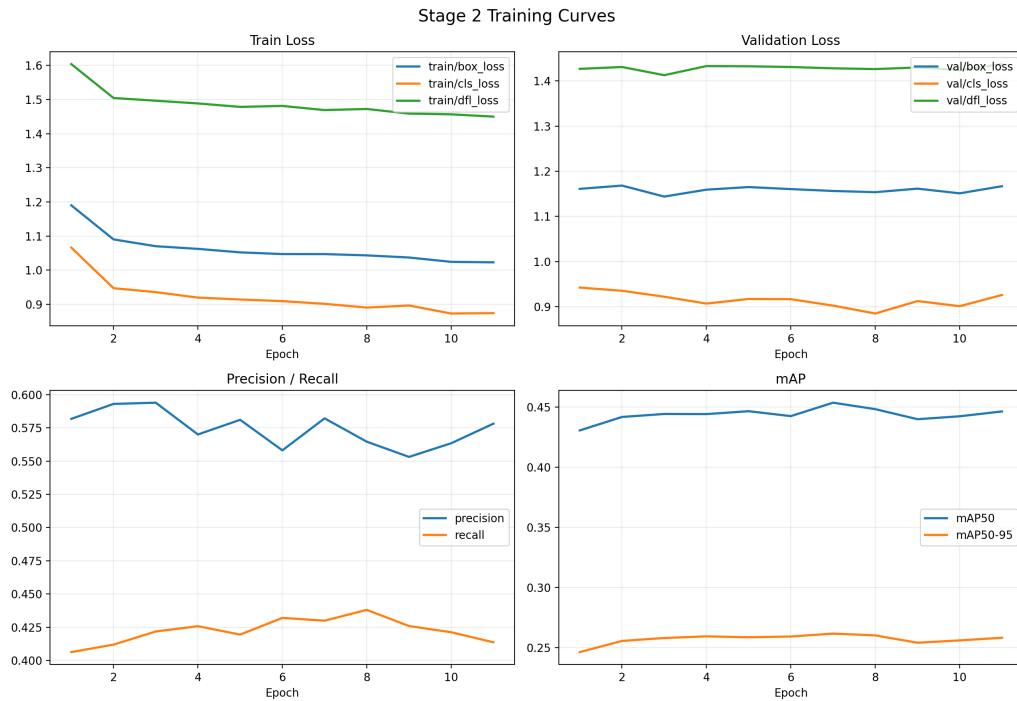


Figure 5.5: Stage 2 training curves.

5.5 Compute Infrastructure and Cost

The main experiments were run on rented NVIDIA L4 GPU resources. The approximate experimental cost was 1,000,000 VND. This is important because the report is not based on a trivial notebook run: the compute budget produced real model checkpoints, training curves, validation metrics, and error-analysis artifacts.

At the same time, GPU cost does not guarantee high mAP. For fire and smoke detection, model capacity is only one part of the problem. Label noise, hard negatives, small objects, duplicate leakage, and inference post-processing can dominate the final result. Therefore, the cost should be interpreted as the cost of a serious experiment and diagnosis, not as proof that the model must be deployment-ready.

Chapter 6

Evaluation and Error Analysis

6.1 Evaluation Metrics

The final model is evaluated as a one-class detector. The main metrics are precision, recall, F1-score, mAP@50, and mAP@50:95:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

6.2 Validation Per-Group Analysis

Overall mAP hides group-specific weaknesses. Table 6.1 therefore reports per-group validation metrics using the Stage 2 checkpoint.

Table 6.1: Per-group validation error analysis

Group	Images	GT	Precision	Recall	mAP@50	Interpretation
g01 positive_standard	527	1,506	0.551	0.410	0.426	Main bottleneck due to low recall in the largest group.
g02 alley_context	42	105	0.690	0.476	0.566	Better than g01, but sample size is small.
g03 hard_negative	19	0	N/A	N/A	N/A	Requires false-positive analysis instead of mAP.
g04 sahi_small_objects	20	108	0.865	0.714	0.760	Slicing helps small objects, but localization remains weak at high IoU.
g05 ambient_null	233	0	N/A	N/A	N/A	Requires false-positive analysis instead of mAP.

The validation results show that small objects are not the only issue. In fact, g04 has the highest validation mAP@50. The major limitation is g01, where many positive standard scenes are still missed. This suggests that additional label QA and positive data diversity are necessary.

6.3 Full-Image Test Evaluation

The standard test benchmark evaluates the Stage 2 checkpoint on the original full-image test split at image size 1024. This is the most comparable setting to validation because both use the normal YOLO validation protocol rather than a custom slicing and merging pipeline. Ultralytics removed two duplicate labels from one test image during evaluation, so the full-image test contains 1,467 valid evaluated instances.

Table 6.2: Full-image test evaluation with the Stage 2 checkpoint

Group	Images	GT	P	R	mAP50	mAP50-95
Overall	733	1,467	0.552	0.430	0.421	0.240
g01 positive_standard	452	1,243	0.547	0.422	0.413	0.246
g02 alley_context	42	132	0.660	0.477	0.561	0.346
g03 hard_negative	19	0	N/A	N/A	N/A	N/A
g04 sahi_small_objects	20	92	0.770	0.543	0.511	0.182
g05 ambient_null	200	0	N/A	N/A	N/A	N/A

The full-image test shows that g01 remains the largest source of missed positives. g02 is the strongest positive test group, while g04 confirms that the detector can still recover many small objects under normal full-frame inference. Because g03 and g05 contain no positive labels, their quality should be measured by false positives per image rather than by mAP.

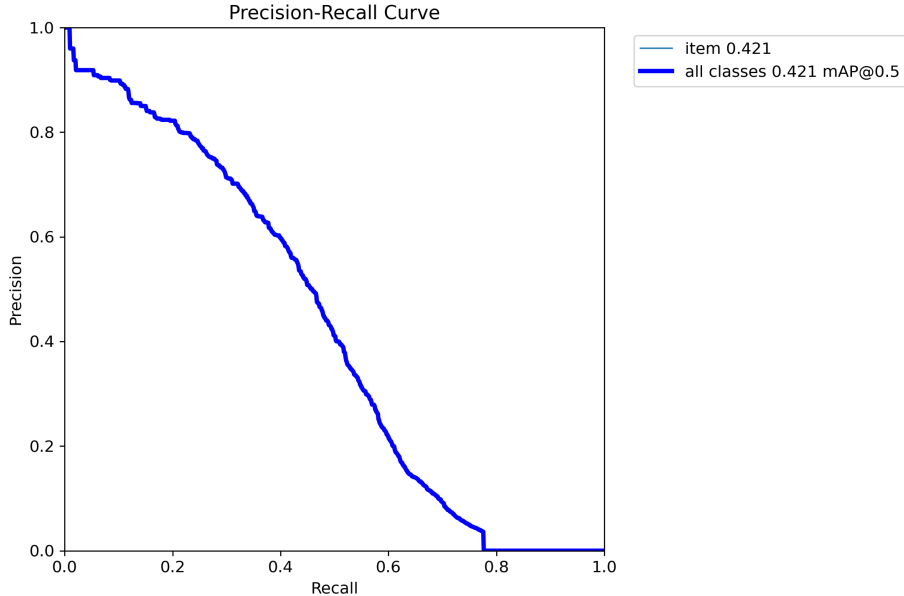


Figure 6.1: Full-image test precision-recall curve for the Stage 2 checkpoint.

The full-image test result changes the interpretation of the experiment. Validation $\text{mAP}@50 = 0.454$ and full-image test $\text{mAP}@50 = 0.421$ are reasonably close, so the model is not collapsing purely because of test-set distribution shift. The much lower sliced result should therefore be treated as an inference-protocol problem until further threshold and NMS sweeps prove otherwise.

6.4 Sliced Inference Ablation

The project also evaluates the Stage 2 checkpoint with sliced inference on the same test split. This protocol uses image size 1024, slice size 768, overlap 0.25, includes full-image prediction,

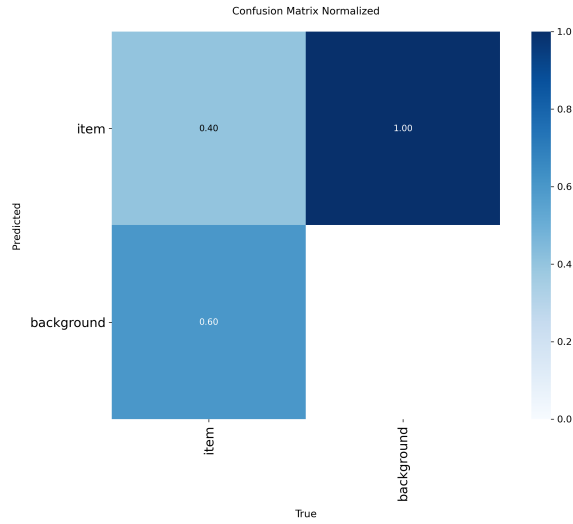


Figure 6.2: Normalized confusion matrix from the full-image test run.

and reports precision/recall at confidence 0.25.

This experiment is useful as an ablation, not as the main test result. Compared with the full-image mAP@50 of 0.421, sliced inference reaches only 0.152. The result shows that the current slicing, merging, and reporting thresholds are harmful in their present configuration, especially for recall.

Table 6.3: Sliced inference evaluation on the test split

Group	Images	GT	mAP@50	TP	FP	FN	Recall@0.25
Overall	733	1,469	0.152	210	292	1,259	0.143
g01	452	1,245	0.202	176	151	1,069	0.141
g02	42	132	0.252	27	35	105	0.205
g03	19	0	N/A	0	6	0	N/A
g04	20	92	0.110	7	2	85	0.076
g05	200	0	N/A	0	98	0	N/A

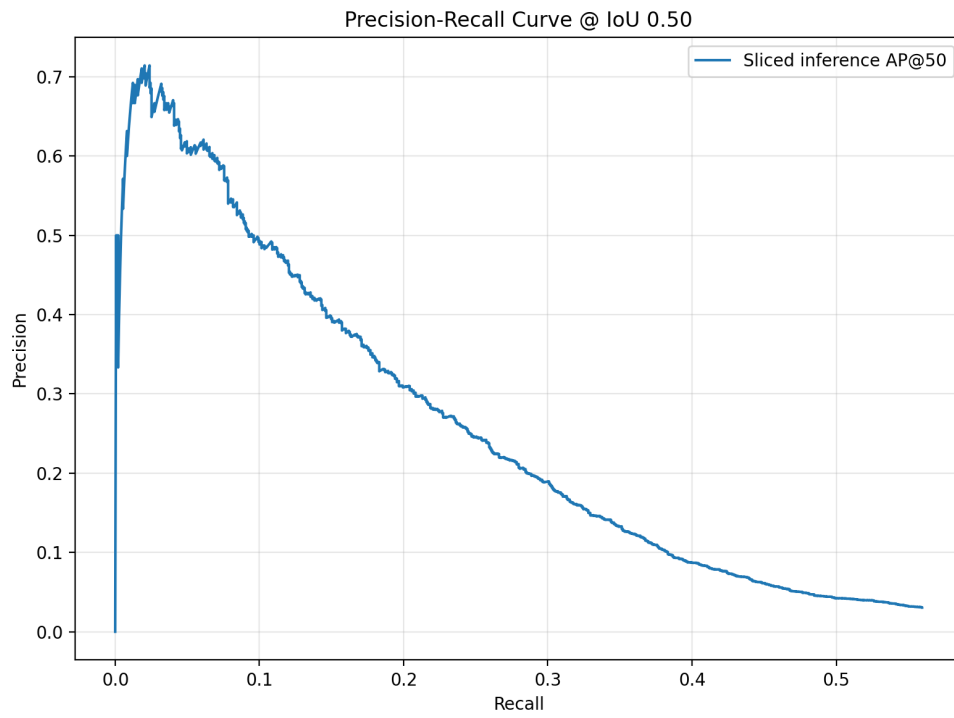


Figure 6.3: Sliced test precision-recall curve at IoU 0.50.



Figure 6.4: Sliced inference prediction example from g01.



Figure 6.5: Sliced inference prediction example with difficult localization.

6.5 Interpreting the Low Metrics

The results are below the original target, but the conclusion should be precise. The current model is not a failed experiment; it is a useful diagnostic experiment. There are three reasons.

1. **Full-image and sliced tests measure different protocols.** Stage 2 validation reaches $\text{mAP}@50 = 0.454$, while the full-image test reaches $\text{mAP}@50 = 0.421$. This is a moderate and credible drop. The much lower sliced $\text{mAP}@50 = 0.152$ points to slicing, merging, and thresholding as likely causes, but a confidence/NMS sweep is still required before assigning exact responsibility.
2. **The dataset is deliberately difficult and leakage-controlled.** Removing near-duplicates and preserving hard negatives can lower the metric compared with an easier leaked benchmark, but it makes the evaluation more honest.

3. **The error analysis points to actionable fixes.** The main bottlenecks are false negatives in g01, low recall after thresholding/merging slices, and false alarms in ambient null scenes.

6.6 Critical Experimental Gaps

6.6.1 Full-Image Baseline Added, Ablations Still Missing

The standard full-image test baseline has now been added, which fixes the largest comparability problem in the earlier analysis. The remaining experimental gap is systematic ablation. A stronger evaluation should still repeat the comparison under controlled threshold and NMS settings:

1. keep full-image YOLO validation as the official test benchmark;
2. run sliced inference using the same checkpoint and test split;
3. sweep report confidence, tile confidence, merge IoU, and NMS settings;
4. report per-group precision, recall, mAP, and false positives per image.

Only this sequence can prove whether slicing improves small-object detection or damages recall through merging and thresholding.

6.6.2 g04 Small-Object Protocol Gap

The g04 sahi_small_objects result is still important, but the full-image test makes the issue clearer. On validation, g04 achieves $\text{mAP}@50 = 0.760$ and $\text{recall} = 0.714$. On full-image test evaluation, g04 remains usable with $\text{mAP}@50 = 0.511$ and $\text{recall} = 0.543$. In sliced test evaluation, however, g04 $\text{recall}@0.25$ drops to 0.076. This means the model can detect some small objects, while the current sliced inference protocol is failing to preserve them.

Several hypotheses are possible:

- the validation and test g04 subsets may differ in object scale or visual difficulty;
- mAP on full-image validation and $\text{recall}@0.25$ on sliced inference are different measurements;
- sliced inference may generate candidate boxes that are removed or merged incorrectly;
- the threshold 0.25 may be too high for small objects;
- the validation subset may be too small to represent true small-object performance.

The report therefore treats this as a protocol gap rather than a solved error. A confidence sweep and a full-image-vs-sliced comparison on g04 are required before concluding that SAHI helps or hurts small-object detection.

6.6.3 No Confidence/NMS Sweep

The current sliced test uses a fixed report confidence of 0.25, tile IoU 0.70, and merge IoU 0.55. Because no sweep is available, the claim that post-processing is the cause of the gap remains a hypothesis. A stronger experiment would evaluate:

$$conf \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35\}$$

and multiple merge-IoU/NMS values. The expected output should be a table or curve showing the precision–recall trade-off, especially for g04 and negative groups.

6.7 Main Error Categories

6.7.1 False Negatives in Positive Standard Scenes

g01 `positive_standard` contributes most of the missed objects. This suggests that the model still fails on common positive cases, not only on rare edge cases. Possible causes include label inconsistency, visually subtle smoke, small flame regions, and large intra-class variation.

6.7.2 Hard Negative Confusion

Hard negatives and ambient null frames remain essential. The test sliced protocol produced false positives in g03 and g05. In a fire alerting system, even a modest false positive rate is harmful because repeated false alarms reduce user trust.

6.7.3 Sliced Inference Post-Processing

Sliced inference was expected to help small objects, but the current configuration produces low recall. The likely issue is not simply model capacity; it may be the interaction among low candidate threshold (`conf=0.001`), merge IoU, NMS, and final report confidence. However, this remains unproven until a sweep is run. A threshold/NMS sweep is required before this can be considered deployment-ready.

Chapter 7

Discussion

7.1 Why the Project Still Has Academic Value

The final metric does not reach the target, but the project demonstrates a rigorous Data Mining process. A superficial report could hide low performance by using duplicated data, easy images, or only overall mAP. This report instead shows the difficult parts: manual labeling, data cleaning, negative sampling, group-wise evaluation, and post-processing diagnosis.

From an academic perspective, this is valuable because it produces falsifiable conclusions:

- The dataset after deduplication is harder but more realistic.
- Stage 2 fine-tuning improves validation mAP only slightly.
- Small-object validation is not the sole bottleneck.
- Full-image test evaluation is the main benchmark, while sliced inference currently reduces recall.
- More GPU compute alone is unlikely to solve the problem without better data and post-processing.

7.2 Recommended Improvement Plan

The next iteration should prioritize data and evaluation rather than blindly training more stages.

1. **Keep full-image test as the official benchmark:** Use the Stage 2 checkpoint result on the original full-image test split as the primary score.
2. **Compare full-image and sliced inference systematically:** Run both protocols on the same split, same checkpoint, and same metric script where possible.

3. **QA the largest false-negative group:** Recheck g01 labels and inspect images where the model misses objects.
4. **Expand hard negatives:** Add more lamps, reflections, sunset, welding, haze, and night-scene backgrounds.
5. **Run confidence/NMS sweeps:** Evaluate `report_conf` from 0.05 to 0.35 and adjust merge IoU for sliced inference.
6. **Quantify annotation quality:** Estimate inter-annotator agreement and label error rate on a sampled subset.
7. **Active learning:** Run inference on unlabeled images and select uncertain samples for annotation.
8. **Repeat L4 experiments after data fixes:** Reuse the same L4 protocol after label QA and post-processing sweeps so improvements remain comparable to the reported experiments.

Chapter 8

Conclusion

This project built a serious Computer Vision and Data Mining pipeline for fire and smoke detection. The strongest contributions are not limited to the final YOLO11x checkpoint. The project collected multi-source data, created segmentation annotations from image-only sources, designed meaningful data groups, removed near-duplicates, prevented leakage through source-level splitting, generated data audit artifacts, trained on rented NVIDIA L4 compute, packaged the Stage 1/2 checkpoints on Hugging Face, implemented a local Streamlit inference interface, and performed group-wise and sliced inference error analysis.

The best Stage 2 validation result achieved $\text{mAP@50} = 0.454$, below the initial target of $\text{mAP@50} \geq 0.70$. The full-image test benchmark achieved $\text{Precision} = 0.552$, $\text{Recall} = 0.430$, $\text{mAP@50} = 0.421$, and $\text{mAP@50:95} = 0.240$. This is the main test result. The sliced test ablation reached $\text{mAP@50} = 0.152$ and $\text{recall@0.25} = 0.143$, showing that the current sliced inference configuration is not ready for deployment.

The main lesson is that fire/smoke detection is a data-centric problem. Compute is necessary, and nearly 1,000,000 VND was spent on rented NVIDIA L4 experiments, but GPU time alone cannot overcome noisy labels, missing annotation-quality statistics, or untuned inference protocols. The correct next step is not merely to train longer, but to improve labels, add targeted hard negatives, sweep threshold/NMS settings, and use active learning to close the recall gap.

Final Statement

Although the final metric is not high, the project is valuable as a rigorous experimental pipeline. It honestly documents the work required to build a dataset, train a modern detector, pay for real GPU experiments, package a checkpoint artifact, provide a usable inference test UI, and diagnose why the model underperforms. The remaining weaknesses are now explicit: missing confidence/NMS sweeps, missing quantitative annotation-quality measurement, background false-positive measurement, and a small-object sliced-inference protocol gap. This makes the report stronger than a simple high-score demo because it shows both engineering capability and

scientific honesty.